

A NEW APPROACH TO IDENTIFYING KEY FINANCIAL MATERIALITY DRIVERS FROM SENTIMENT DATA

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Abstract

This chapter presents a novel, data-driven methodology for identifying financially material environmental, social and governance (ESG) factors or *drivers* through rigorous statistical analysis of sentiment-return relationships. We reduce 26 ESG drivers, aligned to Sustainability Accounting Standards Board (SASB), to 6 material candidates using a two-stage methodology: statistical screening via Mutual Information and Binary Area Under Curve are used to reduce the number of drivers from 26 to 12 and then Random Forest refinement is used to reduce 12 drivers to 6, while maintaining balanced representation across E, S and G pillars.

The selected drivers are validated across rolling 26-week windows to assess stability across market regimes. To authors' knowledge, this is the first and the only systematic as well as transparent ESG driver ranking methodology based on open source data in the literature. Given that over 5,000 companies in the EU and thousands of other companies in the developed world require financial materiality disclosure from 2026 onwards, the authors feel that this is an important step in establishing a global standard on choosing the ESG factors to report on.

1 Introduction

Various ESG drivers materially affect the financial performance of any company, ranging from air quality to supply chain management. However, measuring the financial materiality of ESG factors remains methodologically challenging. There is a strong regulatory push towards standardised reporting of financial materiality of ESG drivers in corporate disclosure. In particular, two major regulatory frameworks now mandate systematic identification and reporting of financially material sustainability drivers:

- **European Union - CSRD/ESRS Framework:** The revised Corporate Sustainability Reporting Directive (CSRD), effective from 2026, requires approximately 5,000 large EU companies to report under the European Sustainability Reporting Standards (ESRS). ESRS explicitly requires double materiality assessment - both financial materiality (sustainability matters affecting enterprise value) and impact materiality (company impacts on society and environment). ESRS 1 mandates that companies identify material sustainability matters through systematic assessment, with financial materiality defined as information relevant to investment decisions. Our pipeline's ability to quantify which ESG drivers correlate with financial performance directly supports CSRD compliance by providing evidence-based materiality rankings.

- **Global International Financial Reporting Standards (IFRS) framework:** The International Sustainability Standards Board (ISSB) standards, adopted in the UK, Australia, Canada, Japan, and being implemented in India, focus explicitly on investor-relevant sustainability information. These standards require over 100,000 companies to disclose material sustainability-related risks and opportunities that could reasonably affect enterprise value, emphasizing quantitative analysis to support materiality determinations. Our methodology’s temporal validation across market regimes and cross-model consensus ranking provides the robust quantitative foundation required for ISSB compliance.

These regulatory frameworks transform ESG materiality from voluntary best practice to mandatory compliance requirement, creating an urgent need for systematic, quantifiable approaches like the one presented here.

1.1 Market Scale and Capital Flows

Capital flows reflect this maturation from voluntary principle to regulated market reality. The EU Taxonomy that defines which economic activities qualify as environmentally sustainable, operationalizes regulatory disclosure into actionable investment criteria. Within the EU, taxonomy-aligned capital expenditure from large listed European firms reached 250 billion in 2023 [1], alongside an estimated 200 billion allocated to credible transition plans [2]. Globally, the green debt market stood at approximately 1.7 trillion in outstanding value at the end of 2023 [3].

Institutional investors reinforce this trend by applying stricter ESG criteria and reallocating capital towards sustainable assets. Union Investment removed ExxonMobil from portfolios for failing climate targets, while Norway’s sovereign wealth fund divested from 49 companies and tightened ESG requirements across 8,700 holdings. Although ESG has attracted political criticism, particularly in the United States, market data shows resilience: ESG-labeled investments experienced only a temporary dip in 2022, with flows and valuations subsequently recovering due to investor conviction and strong liquidity.

1.2 Challenges in Measuring ESG Financial Impact

Assessing the financial materiality of ESG factors presents several methodological challenges. The secondary market introduces a fundamental attribution problem: trades in existing securities do not channel new capital to projects, making it difficult to establish direct causal links between investor behavior and real-world ESG outcomes. Data heterogeneity remains a significant obstacle, as ESG metrics are often self-reported, inconsistent across providers, and susceptible to methodological biases. Temporal dynamics complicates analysis, as market responses to ESG events may be immediate in some cases but lagged in others. Finally, sentiment extraction from unstructured text introduces the risk of noise, misclassification, and spurious correlations if not properly validated.

In the secondary market, when companies report ESG-related activities - reducing emissions, improving labor practices, strengthening governance - investors interpret this information and incorporate it into valuations. Stock prices become market-reflected signals of the perceived financial materiality of those ESG actions. Our approach systematically maps these effects by identifying which specific ESG drivers influence stock prices, correlating structured ESG sentiment with share price movements across multiple time windows to quantify how investors value particular ESG factors.

1.3 Our Approach and Contributions

This research presents a systematic pipeline for ESG sentiment analysis and driver selection, addressing the critical challenge of identifying which sustainability drivers demonstrate robust

statistical relationships with equity performance. Our methodology assesses financial materiality by analyzing the mutual dependence between weekly ESG sentiment scores for individual drivers and share price movements, under varying market conditions.

This framework leverages efficient market principles: when companies disclose ESG activities, investors interpret this information and incorporate it into valuations [4]. Stock prices thus reflect collective market judgment about which sustainability factors affect enterprise value. By systematically analyzing ESG sentiment-return correlations, we identify which drivers the market deems financially material. ¹.

1.4 Case Study Entities

To demonstrate the pipeline’s practical application, we analyze two publicly traded companies representing distinct sectors and ESG profiles:

BP plc: A global energy company operating in oil and gas exploration, refining, and renewable energy transition. BP faces significant Environmental pillar materiality (carbon emissions, energy transition) alongside Social concerns (community relations, safety) and Governance challenges (climate strategy credibility). The 274-week analysis period (January 2020 - March 2025) captures critical regime shifts including the COVID-19 pandemic, green recovery momentum, and the 2022 energy crisis.

Schneider Electric SE: A multinational corporation specializing in energy management and industrial automation, with strong ESG credentials across all three pillars. Schneider Electric provides a contrasting case study to BP, demonstrating how the pipeline performs for companies with relatively stable, high-quality ESG performance and lower sectoral controversy.

Both companies satisfy our heuristically chosen data quality requirements: more than 100 weeks of historical data, more than 15 active ESG drivers per week, and a balanced pillar representation. The chosen data quality requirements can be justified as follows. The 100-week minimum ensures adequate feature-to-sample ratio for machine learning stability. With up to 26 ESG drivers as features, standard ML practice recommends at least 4-5 samples per feature, yielding a minimum of 104-130 weeks. We set the threshold at 100 weeks, which also corresponds to approximately 2 years of data - the period when Mettle Capital’s open source intelligence (OSINT) coverage reached maturity (post-2020). Entities with shorter histories exhibit insufficient ESG discourse volume for reliable sentiment extraction. The 15-driver threshold represents a permissive data completeness filter, rather than a statistical constraint. This ensures sufficient ESG pillar representation (minimum 5 drivers per each of E, S and G pillars) while allowing the subsequent two-stage selection process (Sections 5.2-5.3) to identify a truly material subset. Entities with a small number of active drivers may exhibit insufficient thematic coverage for balanced ESG analysis. Both these choices were established through trial and error, as are a few other choices in latter sections. Comparative analysis of these entities validates the methodology’s robustness across different ESG risk profiles and market conditions.

The rest of the chapter is organized in the following sections.

- The next section outlines ESG Alternative Data Landscape, in terms of characteristics of ESG sentiment data.
- Section 3 deals with the preliminary processing of the raw sentiment data.
- Section 4 discusses our core methodology to rank financial material drivers using freely available machine learning tools, and reduce the number of drivers from 26 to 6.

¹Replication materials, including code, data processing pipelines, and validation outputs for BP and Schneider Electric, are available at: https://github.com/MajidJangani/financial_materiality_sentiment_data

- In section 5, additional insights are provided using temporal validation of our ranking algorithm over a number of time windows.
- Finally, the last section outlines summary of contribution and ongoing research.

2 ESG Sentiment Characteristics

ESG sentiment data provides complementary but distinct information compared to traditional ESG ratings from data providers such as MSCI, Sustainalytics, or Refinitiv. Traditional ratings rely on corporate disclosures, surveys, and standardized metrics, updating quarterly or annually with significant temporal lags [5]. These ratings exhibit low cross-provider correlation and limited predictive power for financial outcomes, partly due to backward-looking methodology and reliance on self-reported data [6]. In contrast, sentiment analysis captures real-time market and stakeholder perceptions of ESG performance, potentially identifying material issues before they appear in corporate disclosures or rating updates.

ESG sentiment differs from general financial news sentiment in both content and temporal dynamics. General financial news focuses on price-relevant information with immediate market implications- earnings results, management changes, competitive developments. ESG sentiment encompasses longer-term sustainability themes that may lack immediate price relevance but carry significant future cash flow implications through regulatory compliance, operational efficiency, or reputation effects [7].

The temporal dynamics differ substantially. General news sentiment exhibits rapid mean reversion as markets incorporate information efficiently. ESG sentiment may persist for extended periods, reflecting structural sustainability challenges or opportunities that require time to manifest in financial performance. This persistence suggests different optimal holding periods for ESG-based strategies compared to news sentiment approaches.

ESG discourse occurs primarily in institutional contexts - regulatory filings, sustainability reports, analyst research, NGO publications - with lower frequency but higher information content per signal compared to social media sentiment [8]. The stakeholder diversity in ESG discourse spans regulators, institutional investors, sustainability professionals, and civil society organizations, each with different expertise levels and time horizons, providing richer information but requiring careful aggregation methods.

Fundamental ESG metrics - carbon emissions, employee turnover, board diversity- provide direct measures of sustainability performance but suffer from reporting lags, limited standardization, and gaming concerns [9]. Sentiment analysis captures market perceptions of ESG performance that incorporate both disclosed metrics and qualitative assessments of management commitment, strategy credibility, and relative positioning. Divergences between fundamental metrics and sentiment may identify opportunities for further investigation or contrarian positioning.

3 Input Data Structure

3.1 The Role of Input Data in ESG Analytics

The strength of any predictive pipeline depends fundamentally on the quality and structure of its input data. For ESG (Environmental, Social, Governance) financial materiality analysis, it is essential to capture both external discourse, reflecting how the world perceives a company, and market behavior, which indicates how stock prices respond. In this pipeline, the foundation rests on two complementary streams: machine-read ESG sentiment data, derived from Mettle Capitals open-source intelligence (OSINT)-based approach, and financial market data price series, from reliable sources such as Yahoo Finance. These inputs allow the transformation of raw

discourse and perception into quantified ESG sentiment and, ultimately, financial materiality insights.

3.2 ESG Driver Taxonomy and Alignment

The ESG sentiment data is organized according to a taxonomy adapted from the *SASB Materiality Map*. Each driver represents a distinct dimension of ESG discourse and is classified under one of three lenses: Environment, Social, or Governance. Table 1 summarizes the full taxonomy along with the number of drivers per lens.

Lens	Drivers	Count
Environment	1. Air Quality, 2. Ecological Impacts, 3. Energy Management, 4. GHG Emissions, 5. Waste & Hazardous Materials Management, 6. Water & Wastewater Management	6
Social	7. Access & Affordability, 8. Customer Privacy, 9. Customer Welfare, 10. Data Security, 11. Employee Engagement, 12. Employee Health & Safety, 13. Human Rights & Community Relations, 14. Labour Practices, 15. Product Quality & Safety, 16. Selling Practices & Product Labelling	10
Governance	17. Business Ethics, 18. Business Model Resilience, 19. Competitive Behaviour, 20. Critical Incident Risk Management, 21. Management of the Legal & Regulatory Environment, 22. Materials Sourcing & Efficiency, 23. Product Design & Lifecycle Management, 24. Physical Impacts of Climate Change, 25. Supply Chain Management, 26. System Risk Management	10

Table 1: ESG Driver Taxonomy adapted from the SASB Materiality Map.

3.3 Sentiment Data: From Raw Text to Numeric Scores

Mettle Capital’s OSINT network systematically collects ESG discourse from diverse sources including global and regional newspapers, wire services, trade press, blogs, social media, investor forums, and sector-specific reports. The coverage spans multiple languages (English, German, French, Spanish, Italian, Dutch) with tagging by country, allowing for the detection of localized ESG concerns. This outside-in approach ensures early identification of emerging issues, often before they appear in corporate filings.

All raw text for a given entity and ESG driver is machine-read and assigned boolean sentiment flags: positive, negative, or neutral. These flags are aggregated weekly, providing counts of each sentiment category per driver. The structured dataset is therefore organized by entity (*i.e.* company), ESG driver and then by the number of positive, negative, and neutral occurrences, creating a machine-readable panel for downstream analysis. This sentiment dataset, indexed by date, entity and driver, is provided by Mettle for the analysis carried out in this chapter.

To quantify sentiment while mitigating neutral dilution, we use a polarity-based score (PBS) derived from sentiment:

$$\text{PBS}_{i,j,k} = \frac{\text{positive}_{i,j,k} - \text{negative}_{i,j,k}}{\text{positive}_{i,j,k} + \text{negative}_{i,j,k} + 0.5 \times \text{neutral}_{i,j,k} + \epsilon}$$

where:

- $\text{PBS}_{i,j,k}$ is the polarity-based score for entity i , driver j , and week k
- $\text{positive}_{i,j,k}$ is the total count of positive sentiment items for entity i , driver j in week k

- $\text{negative}_{i,j,k}$ is the total count of negative sentiment items for entity i , driver j in week k
- $\text{neutral}_{i,j,k}$ is the total count of neutral sentiment items for entity i , driver j in week k
- $\epsilon = 1 \times 10^{-8}$ prevents division by zero

For each entity-driver-week combination, PBS scores are calculated using the total counts of positive, negative, and neutral sentiment items published during that complete week. Aggregation occurs weekly to maintain comparability across firms and sectors. Statistical robustness is ensured by excluding drivers with over 60% zero sentiment counts and enforcing a minimum of 15 active drivers per entity per week.

Finally, weekly log returns

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

are collected for each company under study, where P_t is the adjusted closing price at time t . This provides ground-truth signals against which ESG sentiment is validated.

3.4 Data Structuring and Integrity for the Pipeline

The analytical foundation of this study is a structured panel dataset, engineered to be both statistically sound and financially relevant. It integrates OSINT-derived ESG sentiment drivers with financial market variables, covering the period from 1 January 2020 to 31 March 2025. Each row represents a weekly observation for a single entity, containing PBS (Positive-Bound Sentiment) scores for all the 26 ESG drivers alongside log returns, as illustrated in Table 2. As mentioned earlier, a minimum of 100 weeks of historical data per entity is recommended, with at least 15 active drivers required per weekly observation to ensure data completeness.

Before modeling, this integrated dataset underwent a rigorous, multi-stage validation process to ensure quality across three critical dimensions: data integrity (assessing history length and driver completeness), statistical suitability (verifying the class balance and a ratio of the number of features to the number of data samples being no more than around 1/20), and business logic (confirming balanced representation across E, S, and G pillars and the exclusion of non-material sentiment streams).

The result is a clean, structured, and actionable dataset purpose-built for financial analysis. This foundational quality directly underpins the reliability of all downstream analytical outputs, including the materiality scoring, temporal validation, and regime analysis that form the core of this study’s findings.

Date	Log return	pbs_1	...	pbs_26
2022-01-07	0.1416	0.12	...	0.04
2022-01-14	0.0177	0.15	...	0.03
...	...	...	...	...

Table 2: Panel Dataset Structure for ESG and Financial Inputs. Each ESG driver includes weekly sentiment data transformed into PBS scores for BP.

4 Target Variable Engineering for Financial Performance Classification

Financial markets exhibit non-normal return distributions with skewness, heavy tails, and volatility clustering, rendering traditional mean- and standard-deviation-based labeling methods unstable for ESG materiality analysis [10]. Unlike conventional financial data, ESG factors

often influence returns through risk mitigation rather than pure return enhancement, necessitating robust target creation that captures both upside potential and downside protection. Our framework uses Median Absolute Deviation (MAD) and robust skewness measures for decreasing sensitivity to outliers and for ensuring stable classification boundaries across diverse market regimes.

This approach converts continuous stock returns into categorical targets suitable for production-grade algorithms like Classification and Regression Trees (CART). By avoiding overly granular labels, it ensures reliable differentiation and balanced class distributions, preventing overgeneralization in CART and ensemble methods.

The framework supports two classification schemes:

- Binary Classification: Differentiates ESG-driven value creators from baseline performers.
- Three-Class Classification: Identifies Lagging, Stable, and Leading companies.

Both schemes leverage the same MAD-based, skewness-aware engine, dynamically adjusting thresholds to maintain statistical stability and interpretability across market conditions, providing resilient signals for ESG analysis.

4.1 Core Statistical Measure

The Median Absolute Deviation (MAD) is the primary measure of dispersion, defined as:

$$\text{MAD} = \text{median}(|X - \eta|). \quad (1)$$

Here, X_i represents individual return observations, and η is the median of the return distribution X . Unlike standard deviation (σ), which squares deviations from the mean and is sensitive to outliers, MAD uses the median of absolute deviations, achieving a 50% breakdown point for exceptional robustness in volatile markets [11].

To capture distribution asymmetry while resisting outliers in ESG-financial return analysis, we define robust skewness (γ) as:

$$\text{Robust Skewness}(\gamma) = \frac{\mathbb{E}[(X - \eta)^3]}{\text{MAD}^3}. \quad (2)$$

This measure uses the average of cubed deviations from the median in the numerator, which approaches zero for symmetric distributions, and scales by MAD^3 in the denominator, providing stability analogous to σ^3 in classical skewness but robust to outliers. Positive γ values indicate a longer right tail (positive skew), while negative values signal a longer left tail (negative skew). This robust skewness enables dynamic threshold adjustments that respond to genuine distributional changes without distortion from outliers.

For normal distributions, MAD relates to standard deviation via:

$$\text{MAD} = \sigma \cdot \Phi^{-1}(0.75) \approx \sigma \cdot 0.6745 \quad (3)$$

where Φ is the standard normal cumulative distribution function (CDF). This yields a scaling constant:

$$k = \frac{1}{\Phi^{-1}(0.75)} \approx 1.4826 \quad (4)$$

Thus, a robust standard deviation estimator is:

$$\hat{\sigma}_{\text{robust}} = 1.4826 \times \text{MAD} \quad (5)$$

These measures (η , MAD, γ , and $\hat{\sigma}_{\text{robust}}$) form the statistical foundation for ESG classification, ensuring robustness across non-normal market conditions [12]. The 1.4826 scaling factor ensures consistency with standard deviation under normality while maintaining breakdown point robustness in the presence of outliers [11].

4.2 Binary Classification: ESG Value Creation Detection

To identify companies with ESG-driven financial outperformance, this system employs a single MAD-based threshold, categorizing returns into two classes: Class 1 for superior performance, and Class 0 (Baseline Performance) for market-aligned or underperforming returns [10]. This approach leverages the robust statistical measures defined in (1) and (2) to ensure stable classification in non-normal market environments.

The base multiplier λ adjusts dynamically based on the magnitude of robust skewness (γ) from (2):

$$\lambda = \begin{cases} 0.75 & \text{if } |\gamma| \leq 1.5 \\ 0.55 & \text{if } 1.5 < |\gamma| \leq 2.0 \\ 0.45 & \text{if } |\gamma| > 2.0 \end{cases} \quad (6)$$

This yields an initial threshold:

$$\tau_{\text{base}} = \eta + \lambda \cdot \text{MAD} \quad (7)$$

where η and MAD are defined in (1). To further adapt to skewness direction, an adjustment magnitude is calculated as:

$$\alpha = \min(0.08, |\gamma| \times 0.03) \quad (8)$$

The final threshold is:

$$\tau = \begin{cases} \eta + (\lambda - \alpha) \cdot \text{MAD}, & \text{if } \gamma < 0 \\ \eta + (\lambda + \alpha) \cdot \text{MAD}, & \text{if } \gamma \geq 0 \end{cases} \quad (9)$$

Various threshold values in the above equations were arrived at using trial and error, as follows. The multiplier values ($\lambda \in [0.45, 0.75]$) and the skewness thresholds ($|\gamma| = 1.5, 2.0$) were calibrated iteratively across 50 validation entities to balance two competing objectives: (1) maintaining class balance (target: 30-40% per class), and (2) preserving signal differentiation. Values outside this range consistently failed one criterion - smaller multipliers ($\lambda < 0.45$) produced excessive imbalance ($>70\%$ in single class), while larger multipliers ($\lambda > 0.8$) yielded near-uniform distributions with no predictive signal. The skewness adjustment factor (0.025 for three-class, 0.03 for binary) scales linearly with distribution asymmetry but is capped at 0.08 to prevent overcorrection. These parameters represent empirically-derived thresholds rather than theoretical optima, analogous to hyperparameter tuning in machine learning. Alternative values within 10% produce similar results, confirming robustness (see validation in Section 6).

This mechanism contracts the threshold on the fat-tail side (relaxing negative thresholds for left-skewed distributions, $\gamma < 0$, to avoid over-classifying the laggards) and expands it on the thin-tail side (tightening positive thresholds for right-skewed distributions, $\gamma \geq 0$, to ensure meaningful outperformance detection). Apart from the multiplier bounds ($\lambda \in [0.45, 0.75]$) and the adjustment limits ($\alpha \leq 0.08$) which control overfitting, we also use a fallback to the 70th percentile if class balance fails. These safeguards ensure the threshold remains financially meaningful and statistically robust across diverse market regimes, enhancing the reliability of detection of the mutual dependence between the ESG sentiment values and log returns.

4.3 Three-Class Classification for ESG Performance

To capture the full spectrum of ESG-driven financial performance, we categorize returns into three classes: Lagging (0) for underperformance, Stable (1) for market-aligned performance, and Leading (2) for outperformance. This classification leverages a robust, skewness-adaptive framework to handle non-normal return distributions, ensuring stable and interpretable targets for ESG materiality analysis.

For significant skewness ($|\gamma| > 1.5$), we adjust classification thresholds dynamically with an adjustment magnitude:

$$\alpha = \min(0.08, |\gamma| \times 0.025), \quad (10)$$

where the parameter 0.025 is again arrived at by trial and error. Independent multipliers for lower (λ^-) and upper (λ^+) thresholds account for skewness direction:

$$\begin{cases} \lambda^- = \lambda + \alpha, & \lambda^+ = \lambda - \alpha, & \text{if } \gamma < 0 \\ \lambda^- = \lambda - \alpha, & \lambda^+ = \lambda + \alpha, & \text{if } \gamma \geq 0 \end{cases} \quad (11)$$

Final thresholds are defined as:

$$\tau^- = \eta - \lambda^- \cdot \text{MAD}, \quad \tau^+ = \eta + \lambda^+ \cdot \text{MAD} \quad (12)$$

These yield class intervals:

$$\begin{aligned} \text{Lagging (0)} : X_i &< \tau^- \\ \text{Stable (1)} : \tau^- &\leq X_i < \tau^+ \\ \text{Leading (2)} : X_i &\geq \tau^+ \end{aligned} \quad (13)$$

Performance classes (Leading/Stable/Lagging) reflect return-based categories derived from MAD thresholds and are *not* direct measures of ESG quality. ESG sentiment features are used to predict these performance classes, with the mutual dependence strength indicating financial materiality. If a particular ESG driver, say X, exhibits a high dependence with leading performance class, as measured via Mutual Information (MI), Area Under Curve (AUC), and F1 scores, it indicates that investors collectively deem the driver X to be financially material. The accuracy of *prediction* of stock return class is important only to the extent that we can trust the ranking of drivers themselves.

Signal preservation requires that Lagging and Leading classes together constitute at least 30% of observations, ensuring meaningful differentiation. Class size protection triggers a fallback to the 33rd and 67th percentiles if balance fails, and audit logging tracks thresholds, multipliers, and skewness values for transparency. These mechanisms maintain robust, balanced, and interpretable classifications across market regimes. Table 3 demonstrates the class balance obtained for three-class classification of returns for our chosen companies (BP and Schneider Electric), over 2020-2025.

Table 3: Class balance for three classes

Company	Lagging	Stable	Leading
BP	37%	22%	41%
Schneider Electric	36%	23%	41%

The next table summarizes the two classification systems used in this work.

Table 4: Comparative Analysis of Classification Systems

Feature	Binary System	Three-Class System
Primary Goal	Identify outperformers	Spectrum analysis
Core Method	Single MAD-based threshold	Dual MAD-based thresholds
Skewness Adaptation	3-tier; asymmetric shift	6-tier; asymmetric multipliers
Class Balance Control	Fallback to 70th percentile	Fallback to 33rd/67th percentiles
Optimal Use Case	Superior performance screening	Stable core identification

5 Pipeline for ESG Signal Processing

Building on the preprocessing of sentiment data using the SASB taxonomy, we transform raw sentiment inputs into a single variable per sustainability driver, forming the feature set for our

ensemble classification model. We then develop a classification system with binary and three-class schemes, using self-adaptive thresholds to ensure balanced classes. This section outlines a two-stage driver selection process to identify the most material ESG drivers for financial materiality analysis, addressing high-dimensionality challenges while maintaining statistical and conceptual rigor.

5.1 Two-Stage Selection Approach

The two-stage selection methodology ensures comprehensive ESG representation by balancing statistical validity and thematic coverage across Environmental, Social, and Governance pillars. It progressively improves the feature-to-sample ratio (from 26/274 to 6/274), integrates multiple evaluation criteria (statistical dependency, classification power, and non-linear interactions), and optimizes drivers for downstream temporal analysis across diverse market conditions. By prioritizing balanced coverage over purely data-driven rankings, the approach captures the interconnected spectrum of sustainability-driven value creation and risk, avoiding thematic bias that could arise from market or regulatory fluctuations.

Note that we identify financially material drivers based on *cross-sectional importance* averaged across 274 weeks, excluding temporal stability metrics as selection criteria at this point. Later in section 6, we independently assesses *how stable* that materiality is across temporal regimes. This architecture (identification of drivers followed by post-choice analytics) preserves both persistently material drivers (suitable for sustained ESG investment) and regime-sensitive drivers (valuable for tactical communication timing) without conflating the average importance with temporal consistency. Combining these dimensions into a single selection metric would obscure strategically distinct driver profiles that may serve different business purposes.

We apply statistical screening using complementary metrics - MI, which captures any form of statistical dependency, and Binary AUC, which measures practical classification power. This stage is directly connected with MAD-based target creation system, relying on binary targets (ESG Baseline vs. ESG Value Creation). These targets highlight which drivers have the highest mutual dependence with market outperformance. This stage ensures that only drivers with both statistical and practical relevance are retained.

Then we conduct Random Forest-based driver ranking using the three-class target (Lagging / Stable / Leading). This step benefits from the improved feature-to-sample ratio (12/274), allowing the detection of complex non-linear relationships and feature interactions that would not be feasible with the original 26 features. The three-class perspective provides a more detailed evaluation of performance gradations, ensuring that the selected drivers remain effective across the full spectrum of ESG outcomes.

This stage also lays the foundation for temporal analysis later. The final six drivers must demonstrate not only statistical significance in cross-sectional analysis but also temporal stability across different market regimes. This ensures that the selection process identifies drivers with robustness for ongoing temporal validation and regime detection analysis - rather than simply fitting historical data.

5.2 Stage 1: Statistical Screening and Composite Scoring (26 $\mapsto$ 12 Drivers)

Stage 1 of our ESG feature selection process reduces 26 sustainability drivers to 12, balancing statistical robustness with practical relevance for financial materiality analysis. This stage employs a time-series-aware cross-validation framework and dual-metric scoring to ensure the selected drivers capture meaningful ESG-financial relationships while maintaining balanced representation across Environmental, Social, and Governance pillars.

To avoid look-ahead bias inherent in financial data, we implement 3-fold expanding-window cross-validation with strict chronological ordering. Unlike standard k-fold cross-validation which

randomly shuffles data, our approach respects temporal structure, ensuring that future information never influences past predictions. For a dataset of length $T = 274$ weeks, the three folds are structured as follows:

- Fold 1: Training dataset is weeks 1-68, testing dataset is weeks 69-137;
- Fold 2: training dataset is weeks 1-137, testing dataset is weeks 138-206;
- Fold 3: training dataset is weeks 1-206, testing dataset is weeks 207-274.

This structure preserves temporal ordering of returns while still providing multiple train-and-test datasets with different sizes and also providing progressive validation over entire time series. This structuring is achieved by using scikit-learn’s *TimeSeriesSplit* [13] with $n_{splits} = 3$.

Each driver’s importance is evaluated using two complementary metrics, computed independently on the training data within each fold. Firstly, MI quantifies statistical dependency of a factor f with the three-class target (Lagging/Stable/Leading). MI is computed using scikit-learn’s `mutual_info_classif` function [13], which estimates the joint density $p(f, y)$ non-parametrically via k-nearest neighbor approximation (k=3 neighbors) for each class. Mathematically:

$$MI(f, Y_{3class}) = \sum_{y \in \{0,1,2\}} \int p(f, y) \log \frac{p(f, y)}{p(f)p(y)} df,$$

where f is the factor being evaluated. MI measures the reduction in uncertainty about the target class due to sentiment scores from a particular driver or a factor. $MI=0$ indicates statistical independence, whereas a high MI indicates (linear or nonlinear) dependence.

As a second metric, we use Area Under the ROC Curve (AUC), which measures the classification power of a factor f for the binary target (baseline performance vs. superior performance):

$$AUC(f, Y_{binary}) = P(f(X^+) > f(X^-) | Y_{binary})$$

where X^+ denotes superior returns and X^- denotes baseline returns. $AUC = 0.5$ indicates that the chosen ESG factor has no classification power, where as $AUC = 1$ indicates perfect classification. AUC is computed using `roc_auc_score` function [13]. More information about MI and AUC can be found in standard textbooks on machine learning, *e.g.* [14]. To capture the signal strength regardless of direction, we use:

$$AUC_{max}(f) = \max(AUC(f), 1 - AUC(f)).$$

Both these metrics are averaged over three folds to obtain $MI^{av}(f)$ and $AUC_{max}^{av}(f)$, respectively. A composite score for factor ranking is arrived at by a linear combination of normalized version of these two scores:

$$Score(f) = 0.625 \times \frac{MI(f)}{\max_{f'} MI(f')} + 0.375 \times (2(AUC_{max}(f) - 0.5))$$

The weights (0.625 for MI, 0.375 for AUC) prioritize non-linear relationships (MI) while ensuring some visibility towards highly explainable classification measures (AUC). Specifically, high AUC, but low MI represents good classification potential, but a lack of consistent statistical dependence of target onto the drivers. On the other hand, a high MI, but low AUC indicates strong statistical dependence but poor classification potential. A combination of the two measures is a heuristic choice made by the authors, which appears to lead to consistent and intuitively justifiable rankings for all the companies under consideration. Minor changes in these weights (keeping the weight of MI higher than that of AUC) changes the exact ranking of drivers, but tends to choose the same drivers. The top 12 drivers, ranked by this composite score, advance to Stage 2.

To ensure a balanced ESG representation, a dynamic allocation strategy adjusts driver selection based on the target feature set size, as shown in Table 5. The allocations prioritize higher Social capacity due to its broader taxonomy coverage, guarantee minimum representation for all pillars, and scale proportionally, selecting 3 Environmental, 5 Social, and 4 Governance drivers for a target size of 12. This approach ensures a thematic balance for downstream analysis.

Table 5: Target Driver Allocation by Selection Size

Selection Size	Env.	Soc.	Gov.
≤ 8	2	3	3
≤ 12	3	5	4
> 12	4	6	5

5.3 Stage 2: Random Forest Refinement (12 $\mapsto$ 6 Drivers)

Stage 2 employs Random Forest (RF) modeling to refine the 12 drivers identified in Stage 1 down to a final set of 6, capturing the multivariate interactions essential for robust ESG-financial materiality analysis. This approach recognizes that the most material ESG drivers are not necessarily those with the strongest individual signals, but rather those that work most effectively together within an integrated framework. The RF refinement process provides several key advantages over univariate screening: it harnesses ensemble robustness by combining multiple decision trees to generate stable feature importance scores less sensitive to data perturbations; captures interaction effects by identifying synergistic relationships between ESG factors that traditional statistical methods miss; emphasizes practical utility by optimizing for drivers that demonstrably improve classification performance rather than merely showing statistical association; and ensures forward-looking stability by selecting features that maintain importance across different data subsets and tree structures. This two-stage approach - statistical screening followed by multivariate refinement - ensures a balance between methodological rigor and investment relevance, while maintaining thematic balance across Environmental, Social, and Governance pillars and building on the time-series-aware analysis established in Stage 1.

The RF model uses 100 trees with a maximum depth of 5, minimum split size of 15, and leaf size of 8, balancing predictive power with overfitting prevention for 274 observations. Feature importance is calculated as:

$$I_{\text{RF}}(f) = \frac{1}{T} \sum_{t=1}^T \sum_{n \in N_t(f)} \frac{N_n}{N_{\text{total}}} \Delta \text{Gini}(n) \quad (14)$$

where $T = 100$ is the number of trees, $N_t(f)$ is the set of nodes in tree t splitting on feature f , N_n is the number of samples at node n , $N_{\text{total}} = 274$, and $\Delta \text{Gini}(n)$ is the Gini impurity reduction. MI quantifies statistical dependency using k -nearest neighbor density estimation, complementing the RF importance for robust ranking [13]. RF is also implemented using standard Python commands.

The selection logic prioritizes the top 6 drivers based on $I_{\text{RF}}(f)$, ensuring at least one driver per ESG pillar to maintain balance, guided by the allocation strategy from Stage 1 (e.g., 2 Env., 2 Soc., 2 Gov. for a target of 6). Class imbalance is addressed through automatic class weighting, and bootstrap sampling is disabled to maximize signal capture. A fixed random seed ensures reproducibility.

5.4 Final Selected Drivers

Table 6 and 7 present *feature-level importance metrics* used to rank the 6 selected drivers. The F1 scores shown (0.59-0.68) measure each driver’s univariate classification power - the ability

of that single feature alone to predict ESG performance categories using simple median-based thresholds. These univariate scores serve as components in the composite ranking formula but should not be confused with the Random Forest model’s actual predictive performance (0.30-0.37, reported in Table 8 below). The distinction reflects that individual features show stronger univariate signals than can be realized in multivariate out-of-sample prediction - a common finding in ESG sentiment analysis.

The apparent discrepancy between univariate F1 scores (0.59-0.68) and Random Forest out-of-sample F1 (0.30-0.37) in table 8 does not indicate overfitting but reflects fundamentally different evaluation contexts. Univariate F1 measures in-sample classification using each feature’s optimal threshold on the full dataset a retrospective assessment analogous to perfect hindsight trading. Model F1 measures prospective prediction on chronologically unseen data with multivariate features competing for signal, which is closer to a realistic investment scenario. The gap between these metrics quantifies the *information loss from univariate to multivariate contexts* and *temporal degradation from hindsight to foresight*, not model overfitting. This pattern appears consistently in financial alternative data research, where individual signal strength exceeds achievable portfolio-level alpha, see *e.g.* [10].

Composite ranking formula is given by

$$0.30 \times \text{MI} + 0.05 \times \text{Corr} + 0.25 \times \text{AUC} + 0.40 \times \text{Univ. F1}, \quad (15)$$

where Corr is the correlation. The multipliers were arrived by trial and error so that the driver rankings obtained make intuitive sense. Minor changes in the weightings in the formula above, keeping their relative importance the same, causes changes in precise rankings but selects the same drivers.

Table 6: BP Results

ID	driver_name	pillar	mi	corr	auc	uni_fl
3	Energy Management Env	Environmental	0.063	0.103	0.561	0.381
13	Human Rights & Community Rel	Social	0.031	0.105	0.562	0.405
1	Air Quality	Environmental	0.0478	0.044	0.525	0.333
12	Employee H and S	Social	0.055	0.040	0.519	0.146
15	Product Quality & Safety	Social	0.017	0.047	0.527	0.392
24	Physical Impacts of Climate Change	Governance	0.034	0.042	0.523	0.320

Table 7: Schneider Results

ID	driver_name	pillar	mi	corr	auc	uni_fl
12	Employee H & S	Social	0.031	0.135	0.579	0.362
10	Data Security	Social	0.038	0.090	0.553	0.377
8	Customer Privacy	Social	0.036	0.080	0.547	0.377
15	Product Quality & Safety	Social	0.043	0.045	0.526	0.459
19	Competitive Behaviour	Governance	0.042	0.054	0.526	0.381
24	Physical Impacts of Climate Change	Governance	0.038	0.027	0.516	0.426

These 6 drivers are chosen for temporal validation, where their relative rankings will shift across 26-week windows in response to changing market regimes (Section 6). Figure 1 gives more detailed visualization of how the rankings are arrived at.

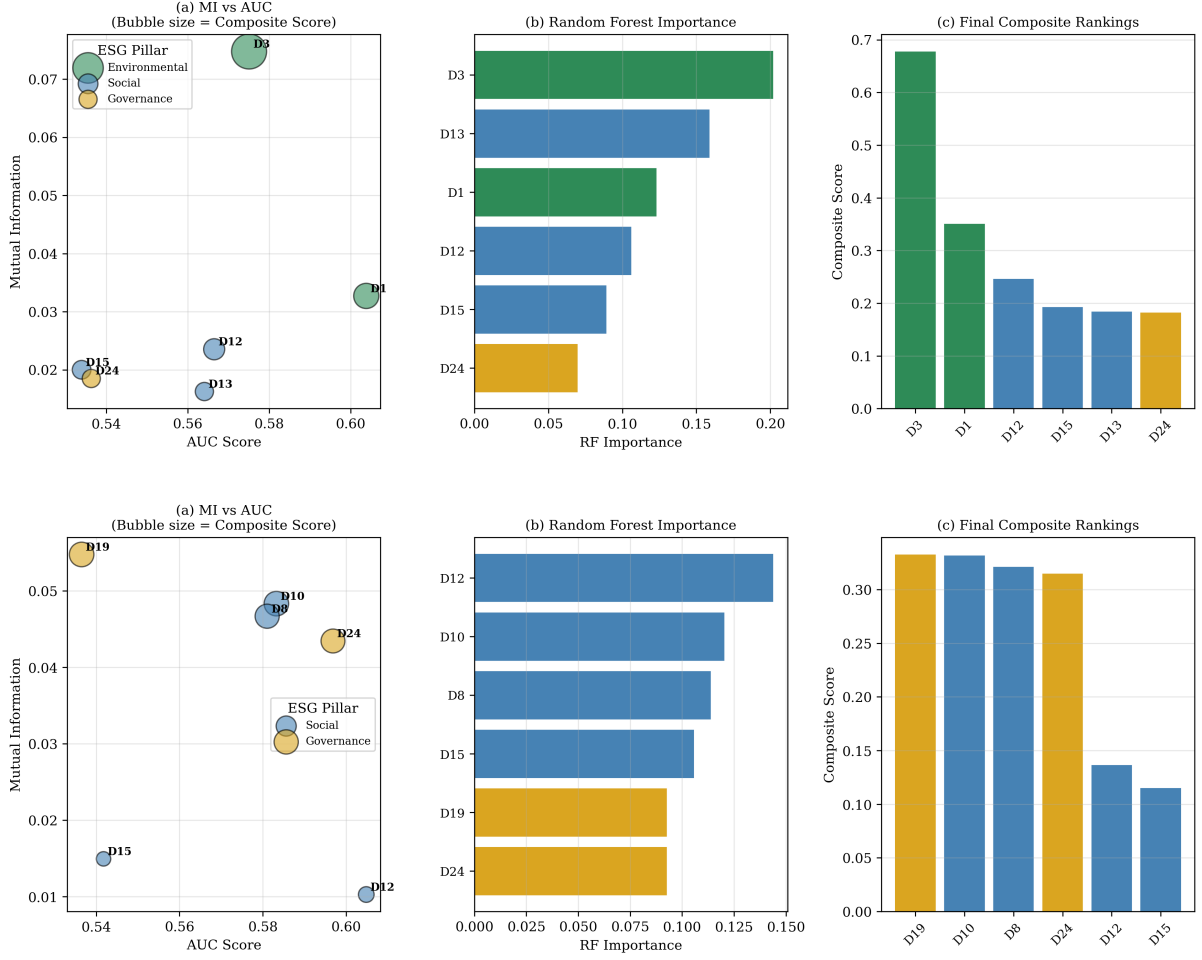


Figure 1: **Top:** BP; **Bottom:** Schneider Electric. Panel (a) plots Mutual Information against AUC scores, with bubble size indicating composite importance. Drivers in the upper-right quadrant demonstrate both high non-linear relationship strength (MI) and strong binary classification power (AUC). Panel (b) shows RF importance scores from Stage 2 refinement. Panel (c) shows composite materiality scores across 17 temporal windows for the 6 selected drivers, where scores are recalculated within each window using equation 15.

Table 8 reports the Random Forest model’s actual predictive performance across the feature selection pipeline, measured via time-series cross-validation on held-out data.

Table 8: Random Forest Classification Performance Across Feature Selection Stages

Entity	All Drivers (26)	Stage 1 (12)	Stage 2 (6)	Improvement
BP	0.299	0.327	0.371	+24.1%
Schneider Electric	0.342	0.336	0.355	+3.8%
Mean	0.321	0.331	0.363	+13.1%

Note: Weighted F1 scores from 3-class classification (Lagging/Stable/Leading) using time-series cross-validation. Scores represent out-of-sample predictive accuracy.

The 13.1% mean improvement demonstrates that dimensionality reduction from 26 to 6 drivers enhances signal-to-noise ratio while eliminating redundant features.

The modest absolute F1 scores (0.30-0.37 range) reflect the inherent challenge of predicting equity returns from ESG sentiment signals alone, consistent with academic literature on

alternative data predictive power [10]. The validation confirms that our two-stage selection successfully identifies signal-bearing drivers, as evidenced by the 24.1% performance gain for BP.

To summarize, the two-stage selection process effectively reduces 26 ESG drivers to 6, balancing statistical rigor with practical utility for financial materiality analysis. Table 9 compares the stages, highlighting their complementary strengths in achieving robust, balanced, and predictive ESG driver selection.

Table 9: Comparison of the Two-Stage Selection Process

Aspect	Stage 1: Statistical Screening (26 $\mapsto$ 12)	Stage 2: RF Refinement (12 $\mapsto$ 6)
Primary Goal	Dimensionality reduction; broad statistical screening	Multivariate refinement; interaction-aware selection
Core Method	Mutual Information + AUC composite scoring	Random Forest feature importance
Target Used	Binary	3-Class
Feature to Sample Ratio	26 : 274	12 : 274
Key Strength	Identifies individual driver significance	Identifies combined effectiveness
ESG Balance	Explicit pillar balancing via allocation strategy	Inherits balance from Stage 1 selection

6 Temporal Validation and Regime Detection

Having identified 6 material ESG drivers through the two-stage selection process (Section 5), we now validate their stability across time using rolling 26-week windows. This analysis tests whether we will get the same drivers across different market conditions, and whether the rankings of these drivers are stable. To have a sufficient number of time windows for tracking, we organize 17 overlapping windows as follows:

- Build rankings for the chosen drivers on weeks 1-26 using the composite score in section 5.4. This gives an ordered list of six materiality drivers.
- Advance time window by one quarter and build driver ranking on weeks 14 - 39, which gives another ordered list.
- Continue in this fashion till the end of data period (February 2025) , with 17th window being the final window of dataset.

This gives us 17 ordered lists of six materiality drivers each, which allows us to analyse the temporal stability of individual drivers as well as E, S and G pillars. We make the following observations.

- Ranking movement of some drivers clearly reflect the changes in the economic or the social regime. As an example, Driver 24 (Physical Climate Impacts) exemplifies regime-dependent materiality through its window-to-window ranking variability:
 1. **Windows 1-5:** Rank 5-6. During the COVID-19-dominated period, climate concerns ranked low as market focus centered on immediate health/safety issues.
 2. **Windows 8-11:** Rank 1-2. Following COP26 momentum and Biden administration climate policy announcements , Physical Climate Impacts ranking improved sharply.

3. **Windows 15-17:** Rank 5-6. As affordability concerns dominated market discourse, climate materiality declined to baseline levels.
- The next two tables give the *average* rank of each driver, along with the volatility of that rank over 17 windows. The first column gives rank over the entire dataset, as obtained in the previous section, so that one can see the variation in rank across smaller (six-month) time windows. It can be seen that the the top two ranks as obtained by the whole data set also lead to the highest average rank. Further, these can intuitively justified for each company. In terms of corporate disclosure, BP needs to focus on energy management and on human rights whereas Schneider Electric being an electricity company needs to focus on employee safety and labour practices. Importantly, our methodology provides a clear, objective and data-driven rationale on which drivers a company should choose to report on and potentially focus on, *e.g.* in terms of capital expenditure or in terms of long term strategic vision.

Table 10: Driver Temporal Stability (BP, 2020-2025)

Overall Rank	Driver	Avg. Rank	Volatility
1	D3 (Energy Mgmt)	1.8	0.6
2	D13 (Human Rights)	2.3	0.7
3	D1 (Air Quality)	3.1	1.2
4	D12 (Employee Safety)	3.4	1.7
5	D15 (Product Quality)	2.9	1.6
6	D24 (Climate)	4.3	1.4

Table 11: Driver Temporal Stability (Schneider Electric, 2020-2025)

Overall Rank	Driver	Avg. Rank	Volatility
1	D12 (Employee Safety)	2.8	1.0
2	D10 (Labor Practices)	3.1	1.7
3	D8 (Engagement)	3.9	1.8
4	D15 (Product Quality)	4.4	1.7
5	D19 (Business Ethics)	3.6	2.0
6	D24 (Climate)	3.2	1.3

7 Conclusion and Ongoing Research

This research presents a comprehensive pipeline for systematic ESG sentiment analysis and driver selection, addressing the critical challenge of identifying which sustainability factors demonstrate robust statistical relationships with equity performance.

The pipeline reduces the initial driver set from 26 to 6 material candidates through a validated two-stage selection process. The selected drivers demonstrate temporal stability across rolling 26-week windows, which provides confirmation for appropriateness of our methodology. The validation confirms that our selection methodology successfully preserves and enhances the modest level of signal which exists in the noisy data. For a company, the analytical outputs represent a financially materiality-related subset of ESG drivers suitable for corporate disclosure and strategic decision making in terms of communications and investment. For an asset manager, the driver rankings for individual companies can help in guiding or assessing investment decisions. The work is continuing to test, fine-tune and extend this methodology for a expanding universe of companies, and developing an industry standard version of this software.

This work contributes to the evolution of ESG investment by providing a systematic framework for processing ESG alternative data. The purpose here is to report on a transparent, valuable and extremely timely methodology with two example applications. This methodology has since been successfully tested on a large number of FTSE 100 companies, with further testing and ‘company ticker to driver rankings’ automation being work in progress.

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